Studybeesofficial

EDUCATIONAL PLATFORM

www.studybeesofficial.com

SSC-I NATIONAL BOOK FOUNDATION FBISE

MATHEMATICS TEST - CHAPTER 1: REAL NUMBERS

Name:		Roll No.		Section:

Total Time: 2:30 Hour **Total Marks: 150**

Time Allowed: 150 Minutes

PAPER OBJECTIVE
SECTION-A (MARKS 150)

NOTE: Section-A is compulsory. All parts of this section are to be answered on these pages and handed over to the center superintendent. Deleting / overwriting is not allowed. Do not use lead pencil.

Q.1 Fill the relevant bubble for each part. All parts carry one mark.

Questions	A	B	C	D	A B C D
1. Which civilization used tally marks on cave walls, bones and stones about 30000 BC?	Romans only	Greeks only	Early humans	Mayans only	○○●○
2. Hippasus named such numbers "irrational", which means	Not whole	Not negative	Not real	Not reasonable	○○○●
3. The notch-marked baboon bone, an early tally record, was discovered in	Iraq	Egypt	South Africa	Italy	○○●○
4. The Babylonian numeral system was a system with base	10	20	60	12	○○●○
5. The Babylonians wrote numerals on clay tablets using	Cuneiform marks	Hieroglyphs	Roman letters	Tally sticks	●○○○
6. Pythagoras used fractions around 500 BC, which led to the discovery of	Irrational numbers	Rational numbers	Integers	Natural numbers	○●○○
7. Romans used the symbols V and X to represent	1 and 5	5 and 10	10 and 50	50 and 100	○●○○
8. Fibonacci published his book on the 0-9 number system in	628	1525	1202	1630	○○●○
9. Sumerians settled near present-day	Egypt	India	Iraq	Greece	○○●○
10. Division of an hour into 60 minutes is a legacy of the	Romans	Greeks	Mayans	Babylonians	○○○●
11. Numbers discovered by early humans for counting are now called	Whole numbers	Natural numbers	Integers	Rational numbers	○●○○
12. The discovery of zero introduced the set of	Whole numbers	Natural numbers	Irrational numbers	Complex numbers	●○○○
13. The word "rational" came from the word	Ratio	Reason	Round	Root	●○○○
14. The hypotenuse of a right isosceles triangle with two equal sides of length 1 unit is	2	√2	1	√3	○●○○

Follow Studybeesofficial on Instagram, YouTube Channel & WhatsApp Channel — Link in Bio

Questions	A	B	C	D	A B C D
15. Al-Khwarizmi called the new digit zero	Zefero	Sifr	Ratio	Attic	○●○○
16. The decimal form of 7/8 is a	Terminating decimal	Recurring decimal	Non-recurring decimal	Irrational decimal	●○○○
17. The set of whole numbers is	{1, 2, 3, ...}	{0, 1, 2, 3, ...}	{..., −1, 0, 1, ...}	{−1, −2, −3, ...}	○●○○
18. The set of natural numbers is	{1, 2, 3, ...}	{0, 1, 2, ...}	{..., −1, 0, 1, ...}	{0}	●○○○
19. Q and Q′ are disjoint sets because	Q ∪ Q′ = R	Q ⊂ Q′	Q ∩ Q′ = ∅	Q = Q′	○○●○
20. Since R = Q ∪ Q′, the sets Q and Q′ are called	Exhaustive sets	Equal sets	Equivalent sets	Overlapping sets	●○○○
21. The distance between two consecutive integers on a number line is the	Unit length	Origin	Coordinate	Radius	●○○○
22. The number −π = −3.14159... lies between	−3 and −2	3 and 4	−5 and −4	−4 and −3	○○○●
23. Which number is a non-integer rational number?	−5	0	7	9/13	○○○●
24. The decimal form of −1/3 is a	Terminating decimal	Recurring decimal	Irrational decimal	Whole number	○●○○
25. The number line is also called the	Complex line	Real line	Integer line	Rational line	○●○○
26. The set Z of integers is	{..., −2, −1, 0, 1, 2, ...}	{0, 1, 2, ...}	{1, 2, 3, ...}	{−1, −2, −3, ...}	●○○○
27. The set of real numbers R equals	Q ∩ Q′	Q − Q′	Q′ − Q	Q ∪ Q′	○○○●
28. Which number is irrational?	23/25	√23	−5	0	○●○○
29. The intersection of rational numbers Q and irrational numbers Q′ is	∅	R	Q	Z	●○○○
30. The decimal form of 5/8 is	0.725	0.5	0.875	0.625	○○○●
31. The decimal form of 2/3 is	0.888...	0.8333...	0.666...	0.1666...	○○●○

Follow Studybeesofficial on Instagram, YouTube Channel & WhatsApp Channel — Link in Bio

Questions	A	B	C	D	A B C D
32. Which one is irrational: √15, √4, √25 or √36?	√4	√15	√36	√25	○●○○
33. Which one is irrational: √9, √2, √49 or √100?	√9	√100	√49	√2	○○○●
34. The decimal form of 7/20 is	0.6	0.225	0.45	0.35	○○○●
35. The decimal form of 3/4 is	1.0	0.75	0.7	0.8	○●○○
36. Which one is irrational: √64, √121, √3 or √4?	√4	√121	√64	√3	○○○●
37. The symbols < and > were introduced by Thomas Harriot around	1525	1630	1202	628	○●○○
38. (0.2)(5) = 1 shows that 5 is the multiplicative inverse of	5	1	0.2	0	○○●○
39. For real numbers a and b, only one of a < b, a = b, a > b is true. This is the	Transitive property	Closure property	Reflexive property	Trichotomy property	○○○●
40. Which real number is its own multiplicative inverse?	0 and 1	Only 0	1 and −1	Only 2	○○●○
41. (x + 5) − 1 = x + (5 − 1) is justified by the	Associative property	Commutative property	Distributive property	Inverse property	●○○○
42. If a > b and c < 0, then	ac > bc	ac = bc	a + c = b	ac < bc	○○○●
43. If a = b then b = a is the	Reflexive property	Transitive property	Multiplicative property	Symmetric property	○○○●
44. A statement using symbols like < or > is called an	Equality	Inequality	Identity	Equation	○●○○
45. By the reflexive property, $y^2 - 17$ equals	$y^2 + 17$	$y - 17$	$y^2 - 17$	$-17 - y^2$	○○●○
46. a(b + c − d) equals	ab + ac + ad	ab + ac − ad	ab − ac − ad	ab + c − d	○●○○
47. If a < b and b < c, then	a > c	a < c	a = c	c < a	○●○○
48. The additive inverse of a is	1/a	a	0	−a	○○○●

Questions	A	B	C	D	A B C D
49. −(3b) + (3b) = 0 is justified by the	Additive identity	Commutative property	Additive inverse property	Closure property	○○●○
50. If 2 < 5 and 5 < 9, then 2 < 9 uses the	Additive property	Trichotomy property	Reflexive property	Transitive property	○○○●
51. If a > 0, then √a is a	Real number	Integer	Complex number	Natural number always	●○○○
52. The multiplicative identity of real numbers is	0	−1	1	a	○○●○
53. If a = b then a + c = b + c is the	Transitive property	Cancellation property	Additive property of equality	Reflexive property	○○●○
54. Every real number is irrational. This statement is	True	True only for roots	False, e.g. √2	False, e.g. 3	○○○●
55. Every integer is a	Natural number	Whole number	Rational number	Irrational number	○○●○
56. The multiplicative inverse of a (a ≠ 0) is	−a	1/a	a	0	○●○○
57. −3(2 − y) = −6 + 3y is justified by the	Associative property	Closure property	Identity property	Distributive property	○○○●
58. (a + b) + c = a + (b + c) is the	Commutative property	Distributive property	Associative property	Identity property	○○●○
59. Every rational number is an integer. This statement is	True	False, e.g. 1/2	True only for fractions	False, e.g. 5	○●○○
60. The real number having no multiplicative inverse is	1	−1	0	2	○○●○
61. The property a(b + c) = ab + ac is called the	Associative property	Distributive property	Closure property	Commutative property	○●○○
62. Multiplying both sides of an inequality by a positive number	Reverses the sign	Makes it an equality	Removes the sign	Keeps the sign unchanged	○○○●
63. If a + b = a, then b equals	0	1	a	−1	●○○○
64. The additive identity of real numbers is	0	1	−1	a	●○○○
65. Multiplying both sides of an inequality by a negative number	Keeps the sign unchanged	Makes it an equality	Is not allowed	Reverses the sign	○○○●

Follow Studybeesofficial on Instagram, YouTube Channel & WhatsApp Channel — Link in Bio

Questions	A	B	C	D	A B C D
66. For real numbers a and b, a + b ∈ R is the	Commutative property	Associative property	Identity property	Closure property	○○○●
67. If 2x + 3 = y, then y = 2x + 3 uses the	Reflexive property	Symmetric property	Transitive property	Additive property	○●○○
68. The distributive property also holds for multiplication over	Subtraction	Division by zero	Powers	Roots	●○○○
69. If a · b = 1, then b equals	1/a	1	a	−1	●○○○
70. Every irrational number is a	Rational number	Integer	Real number	Whole number	○○●○
71. The graph of x ≤ 2 on a number line has a	Filled circle at 2	Hollow circle at 2	Filled circle at 0	Hollow circle at 0	●○○○
72. If a · b = a (a ≠ 0), then b equals	0	1/a	−1	1	○○○●
73. In the graph of x > −4, the circle at −4 is	Filled	Absent	Hollow	Half-filled	○○●○
74. If 4 < 7 and both sides are multiplied by -5, the result is	-35 > -20	-35 < -20	-20 = -35	-20 > -35	○○○●
75. The additive inverse of 3 is	−3	−1/3	0	3	●○○○
76. The plural of "index" is	Indexes	Indices	Indicis	Indexs	○●○○
77. The two square roots of a positive number are	Equal	Additive inverses	Multiplicative inverses	Both positive	○●○○
78. The index of a cube root radical is	3	2	1	0	●○○○
79. $\sqrt[n]{a}$ / $\sqrt[n]{b}$ equals	$\sqrt[n]{(ab)}$	$\sqrt[n]{a} - \sqrt[n]{b}$	$\sqrt[n]{(a - b)}$	$\sqrt[n]{(a/b)}$	○○○●
80. Any positive real number has how many square roots?	One	Two	Three	Four	○●○○
81. √b is the same as	$\sqrt[3]{b}$	b^2	b/2	$\sqrt[2]{b}$	○○○●
82. In √25, the number 25 is called the	Index	Radicand	Base	Exponent	○●○○

Follow Studybeesofficial on Instagram, YouTube Channel & WhatsApp Channel — Link in Bio

Questions	A	B	C	D	A B C D
83. In $\sqrt[3]{35}$, the radicand is	35	3	3 and 35	1	●○○○
84. The symbol √ is called the	Index	Radical sign	Radicand	Exponent	○●○○
85. In $\sqrt[5]{(xy/z)}$, the index of the radical is	xy	z	1	5	○○○●
86. The radical sign was first used in	1630 AD	1202 AD	1525 AD	628 AD	○○●○
87. The positive square root of a number is called its	Principal square root	Radicand	Index	Base	●○○○
88. Write 5^(4/3) in radical form	5^{12}	$\sqrt[3]{(20)}$	$\sqrt[3]{(5^4)}$	$\sqrt[4]{(5^3)}$	○○●○
89. Simplify $\sqrt[3]{6} \times \sqrt[3]{36}$	12	6	42	5	○●○○
90. Simplify $\sqrt{75} \times \sqrt{3}$	112	78	16	15	○○○●
91. Write 10^(5/3) in radical form	$\sqrt[5]{(10^3)}$	$\sqrt[3]{(50)}$	$\sqrt[5]{(30)}$	$\sqrt[3]{(10^5)}$	○○○●
92. The cube root of 729 is	243	9	18	11	○●○○
93. Using distributive property, 32(1/8 + 1/4) =	8/3	12	24	13	○●○○
94. Simplify $\sqrt{2} \times \sqrt{8}$	8	3	$\sqrt{10}$	4	○○○●
95. The principal square root of 361 is	24	21	19	20	○○●○
96. If a = 12, b = 15, c = 5, find a(b + c)	240	900	87	228	●○○○
97. Write 9^(7/4) in radical form	9^{28}	$\sqrt[7]{(36)}$	$\sqrt[4]{(9^7)}$	$\sqrt[4]{(63)}$	○○●○
98. Write 3^(3/4) in radical form	$\sqrt[3]{(12)}$	$\sqrt[4]{(9)}$	$\sqrt[3]{(3^4)}$	$\sqrt[4]{(3^3)}$	○○○●
99. Write 4^(3/5) in radical form	$\sqrt[3]{(4^5)}$	$\sqrt[5]{(12)}$	$\sqrt[5]{(4^3)}$	4^{15}	○○●○

Follow Studybeesofficial on Instagram, YouTube Channel & WhatsApp Channel — Link in Bio

Questions	A	B	C	D	A B C D
100. Simplify √12 × √3	6	5	12	18	●○○○
101. Simplify ∛25 × ∛5	4	5	10	41	○●○○
102. Simplify √45 / √5	9	2	6	3	○○○●
103. To evaluate b^(m/n), first find the nth root of b and then	Multiply by m	Divide by m	Add m	Raise it to the power m	○○○●
104. In b^(m/n), the number n indicates the	Power to be raised	Base	Index of the radical	Sign	○○●○
105. The negative exponent a^{-n} is defined as	$-a^n$	a^n	$1/a^n$	n/a	○○●○
106. When the exponent is a natural number, the base can be	Only positive	Any real number	Only integers	Only natural	○●○○
107. The rule $(ab)^n = a^n b^n$ is called the	Power of a product rule	Power of a quotient rule	Product rule	Quotient rule	●○○○
108. In the expression 3^4, the number 3 is called the	Exponent	Radicand	Index	Base	○○○●
109. $a^r \cdot a^{-s} \div a^s$ simplifies to	a^{r-2s}	a^{r-s}	a^{r+2s}	a^{rs}	●○○○
110. For $a \neq 0$, a^0 equals	0	a	1	−1	○○●○
111. b^(1/n) is equal to	$\sqrt[n]{b}$	b^n	$n\sqrt{b}$	b/n	●○○○
112. The quotient rule of exponents states that a^m / a^n equals	a^{m-n}	a^{m+n}	a^{mn}	a^{n-m}	●○○○
113. Simplify $(3^2)^3$	3^7	6^3	3^6	3^8	○○●○
114. Simplify $7^7 \times 7^2$	7^{14}	7^5	49^9	7^9	○○○●
115. Evaluate 27^(2/3)	1/9	243	9	27	○○●○
116. Simplify $2^4 \times 2^4$	2^8	4^8	2^9	2^0	●○○○

Study Bees OFFICIAL PAK

Questions	A	B	C	D	A B C D
117. Simplify $a^4 \cdot a^{-6}$	a^{24}	a^{10}	a^{-2}	a^{-10}	○○●○
118. Simplify $(5x)^3$	$125x^3$	$125x^4$	$125x$	$5x^3$	●○○○
119. Evaluate 81^(−1/4)	1/3	1/6	1/81	3	●○○○
120. Evaluate $(1/2)^{-4}$	1/16	2/4	16/1	−1/16	○○●○
121. Evaluate 125^(4/3)	129	625	20	3125	○●○○
122. Simplify (25/49)^(−1/2)	25/49	49/50	7/5	5/7	○○●○
123. Simplify $2^6 \times 2^4$	2^{24}	2^{11}	2^{10}	2^2	○○●○
124. Simplify $3^9 \div 3^4$	3^5	3^{13}	3^{-5}	3^{36}	●○○○
125. The value of 100^0 is	1	Undefined	100	−1	●○○○
126. Evaluate $(3/2)^{-3}$	2/9	8/27	27/8	6/9	○●○○
127. Simplify a^{10} / a^4	a^{14}	a^6	1	a^{-6}	○●○○
128. Simplify $x^6 \cdot x^3$	x^{10}	x^{18}	x^9	x^3	○○●○
129. Simplify $a^5 \cdot a^{-2}$	a^3	a^{-3}	a^{-7}	a^{10}	●○○○
130. Evaluate 27^(−2/3)	−9	9	1/9	1/6	○○●○
131. Evaluate 243^(2/5)	9	6	1/9	486	●○○○
132. Evaluate 6^{-2}	-36	−1/36	1/12	1/36	○○○●
133. Evaluate $(2/3)^{-2}$	−4/9	3/4	4/9	9/4	○○○●

Study Bees OFFICIAL PAK

Questions	A	B	C	D	A B C D
134. The value of 3^0 is	-1	1	3	0	○●○○
135. Simplify $(3^4)^2$	12^2	3^8	3^6	3^9	○●○○
136. Simplify $(y^5)^2$	y^{10}	y^{25}	$2y^5$	y^{11}	●○○○
137. Simplify $(3x)^2$	$9x^2$	$9x$	$9x^3$	$3x^2$	●○○○
138. Evaluate 5^{-2}	1/10	1/7	$-1/25$	1/25	○○○●
139. Ratio, proportion and percentage in finance use	Natural numbers only	Irrational numbers only	Imaginary numbers	Rational numbers	○○○●
140. Find 30% of 4500	150	1350	675	1450	○●○○
141. A sum of Rs. 900 is divided in the ratio 1 : 2 : 3. The smallest share is	Rs. 450	Rs. 300	Rs. 150	Rs. 200	○○●○
142. Qasim had Rs. 1,75,000. He wrote cheques for Rs. 45,790 and Rs. 1,12,921. His balance is	Rs. 18,289	Rs. 16,289	Rs. 15,289	Rs. 16,389	○●○○
143. Wajid drove 283.4 km on 16.2 litres. Average km per litre is about	175	16.2	17.5	27.5	○○●○
144. Convert 50°C to Fahrenheit using F = 9C/5 + 32	122°F	117°F	113°F	127°F	●○○○
145. Momina walks at 3.5 km/h for 12 hours. Distance walked is	42 km	4.2 km	45 km	40 km	●○○○
146. Convert 20°C to Kelvin using K = C + 273	290 K	303 K	393 K	293 K	○○○●
147. Convert 37°C to Kelvin using K = C + 273	310 K	313 K	307 K	300 K	●○○○
148. Find 12% of 5000	500	600	300	416	○●○○
149. Convert 50°C to Kelvin using K = C + 273	333 K	323 K	326 K	423 K	○●○○
150. Convert −10°C to Fahrenheit using F = 9C/5 + 32	19°F	32°F	14°F	23°F	○○●○

StudyBees OFFICIAL PAK